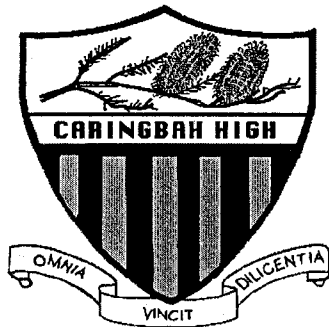


Caringbah High School

YEAR 12 MATHEMATICS SEMESTER 1 EXAMINATION 2009



Time Allowed: 2 hours + 5 minutes reading time

Instructions

- All questions must be attempted.
- Start each question on a new page.
- Answers without necessary working and diagrams may not attract full marks.
- Marks may not be awarded for careless or badly arranged work.
- Diagrams and working which lead to a correct solution may attract some marks.
- Approved calculators may be used

Question 1 (7 Marks)**Marks**

a) Find the primitive function when

i) $\frac{dy}{dx} = 2x^3 - 3$

1

ii) $f'(x) = \frac{4}{x^2}$

1

iii) $\frac{dy}{dx} = \sqrt[3]{x}$

1b) Find the sum of the series $6 + 11 + 16 + \dots + 156$.**2**

c) The sum of the first two terms in an arithmetic sequence is 20. The common difference between successive terms is 4. Find the sum of the first three terms.

2**Question 2 (7 Marks)**a) Find the second derivative of $y = 4x^3 - 2\sqrt{x}$.**2**b) Find the gradient of the curve $y = 5x - x^2$ at the point where $x = 2$.**2**c) State the coordinates of the focus of the parabola $x^2 = 2y$.**1**d) Evaluate $\int_1^3 (3x^4 + 1) dx$ **2****Question 3 (7 Marks)**a) Find $\int \frac{5x^2 + 1}{x^2} dx$ **2**b) i) Sketch, without using calculus, the graph of $y = \sqrt{2x + 1}$, showing intercepts on the x and y axes.**2**ii) Calculate the area bounded by the curve $y = \sqrt{2x + 1}$ and the coordinate axes.**3**

Question 4 (7 Marks)**Marks**

- a) $\int (2x + 5)^3 dx$ 1
- b) For the curve $y = x^3 - 3x^2 + 2$,
- i) Show that stationary points exist on the curve at $x = 0$ and $x = 2$. 2
 - ii) State the nature of each of these stationary points. 2
 - iii) Show that a point of inflection occurs at $(1, 0)$. 1
 - iv) Sketch the graph of $y = x^3 - 3x^2 + 2$. 1

Question 5 (7 Marks)

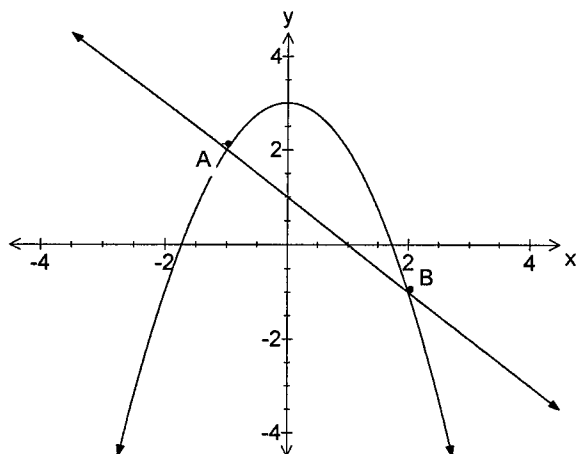
- a) A company invests \$60 000 in equipment. The value of the equipment depreciates by 15% p.a. What is the equipment worth after 10 years? 1
- b) A man planted a tree that was 1.2m tall and it grew 90cm in the first year. Each subsequent year it grew 20% less than it did in the previous year.
- i) How much did it grow in the 5th year after it was planted? 2
 - ii) Calculate the height of the tree 5 years after it was planted. 2
 - iii) Show that the maximum height that the tree can grow is 5.7m 2

Question 6 (7 Marks)

- a) For the parabola $x^2 = 8y - 8$, find
- i) The coordinates of the vertex. 1
 - ii) The coordinates of the focus. 1
 - iii) The equation of the directrix. 1
- b) Find the equation of the locus of a point that moves such that its distance from the point $(2, 3)$ is equal to its perpendicular distance from the line $y = -1$. 2
- c) Find the volume of the solid formed when the area between the curve $y = x^3$, the y -axis and the lines $y = -8$ and $x = 8$ is rotated about the y -axis. 2

Question 7 (7 Marks)**Marks**

- a) Find the equation of the line that is equidistant from the points A(-1,2) and B(5,4). **2**
- b) This diagram shows the line $x + y - 1 = 0$ intersecting with the parabola $y = 3 - x^2$ at the points A and B.



- i) By solving the equations simultaneously, show that the coordinates of A and B are (-1,2) and (2,-1) respectively **2**
- ii) Find the area enclosed by the parabola $y = 3 - x^2$ and the line $x + y - 1 = 0$. **3**

Question 8 (7 Marks)

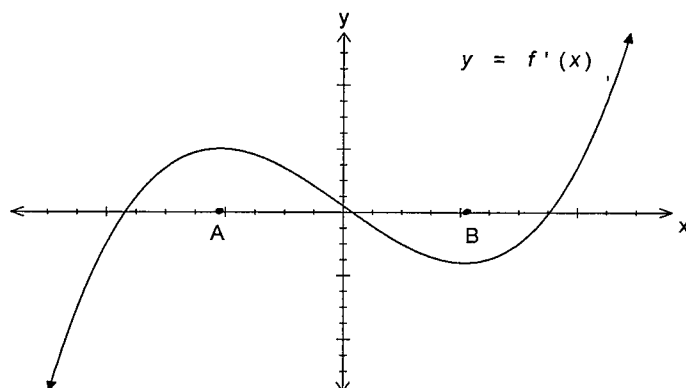
- a) For the function $f(x) = x^3 - 12x$, state the domain where $f(x)$ is increasing. **2**
- b) Two positive numbers have a sum of 24. Find the two numbers if the sum of their squares is to be a minimum. **3**
- c) The following table lists the values of a function for 5 values of x. **2**

x		1.0	1.5	2.0	2.5	3.0
$f(x)$		1.3	1.8	2.1	1.2	0.6

By applying the trapezoidal rule with 5 function values, estimate $\int_1^3 f(x) dx$

Question 9 (8 Marks)**Marks**

- a) Find the equation of the circle with centre (4,-3) and radius 9 units. 1
- b) The sketch below shows the graph of $y = f'(x)$.



- i) Copy the sketch of $y = f'(x)$ on to your answer sheet.
- ii) Explain why points of inflection exist on $y = f(x)$ at $x = A$ and $x = B$. 1
- iii) On the same set of axes as $y = f'(x)$, draw a curve $y = f(x)$ that has the features indicated by $y = f'(x)$. 2
- c) A woman invests \$5000 into a superannuation package at the beginning of each year. The money invested earns 8% p.a. interest. What is the value of the package after 20 years? 3

Question 10 (7 Marks)

- a) Evaluate $\sum_{k=2}^5 2^k - 1$ 1
- b) A local car dealer was offering a loan on the purchase of a \$20 000 car over 5 years. As an incentive to buy, they were advertising “no interest” on the first 6 months. After the first 6 months, interest of $1\frac{1}{2}\%$ per month on the amount owing was charged. A customer agrees to make 60 monthly repayments of \$M, with the first instalment due 1 month after the purchase of the car.
- If A_n is the amount owing after n months, then
- i) Find an expression for A_6 . 1
- ii) Show that $A_8 = (20\,000 - 6M) \times 1.015^2 - M(1 + 1.015)$. 2
- iii) Hence find an expression for A_{60} . 1
- iv) Find the value of M. 2

END OF EXAM

CAS Mathematics Semester 1 2009

Question 1

a) i) $y = \frac{x^4}{2} - 3x + c$

ii) $f(x) = -\frac{4}{x} + c$

iii) $y = \frac{3x^{\frac{4}{3}}}{4} + c$

b) $S_n = \frac{n}{2}(a+1)$
 $= \frac{31}{2}(6+156)$
 $= \underline{2511}$

c) $a + (a+d) = 20$, $d=4$
 $2a = 16$
 $a = 8$

$8 + 12 + 16 = \underline{36}$

Question 2

a) $y = 4x^3 - 2x^{\frac{1}{2}}$
 $y' = 12x^2 - x^{-\frac{1}{2}}$
 $y'' = \underline{24x + \frac{1}{2}x^{-\frac{3}{2}}}$ or $24x + \frac{1}{2\sqrt{x^3}}$

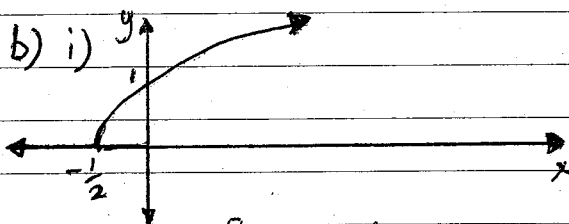
b) $y = 5x - x^2$
 $y' = 5 - 2x$, at $x=2$
 $m = 5 - 4$
 $m = \underline{1}$

c) $x^2 = 2y \therefore a = \frac{1}{2}$
Focus $\underline{S(0, \frac{1}{2})}$

d) $\int_1^3 (3x^4 + 1) dx = \left[\frac{3x^5}{5} + x \right]_1^3$
 $= \left[\left(\frac{3 \times 3^5}{5} + 3 \right) - \left(\frac{3 \times 1^5}{5} + 1 \right) \right]$
 $= \underline{147.2}$

Question 3

a) $\int 5 + x^{-2} dx = \underline{5x - \frac{1}{x} + c}$



ii) $A = \int_{-\frac{1}{2}}^0 (2x+1)^{\frac{1}{2}} dx$
 $= \left[\frac{(2x+1)^{\frac{3}{2}}}{\frac{3}{2} \times 2} \right]_{-\frac{1}{2}}^0$
 $= \left[\frac{1}{3} - 0 \right]$
 $= \underline{\frac{1}{3}}$

Question 4

a) $\int (2x+5)^3 dx = \frac{(2x+5)^4}{8} + c$

b) $y = x^3 - 3x^2 + 2$
 $y' = 3x^2 - 6x$
 $y'' = 6x - 6$

i) Stat pts when $y' = 0$
 $3x^2 - 6x = 0$

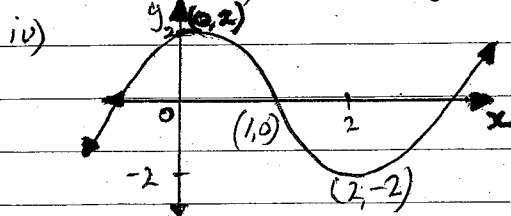
$3x(x-2) = 0 \therefore \underline{x=0}$, $\underline{x=2}$

ii) $x=0$ $y'' = -6 \therefore \underline{\text{max TP at } (0, 2)}$
 $x=2$ $y'' = 6 \therefore \underline{\text{min TP at } (2, -2)}$

iii) Inflection when $y'' = 0$

$6x - 6 = 0 \therefore \underline{x=1}$

From part ii) we can see a change in concavity at $x=1$, $y=0$.



Question 5

a) $A = \$60000 \times (0.85)^{10}$
 $= \$11812.46$

b) i) $T_n = ar^{n-1}$
 $= 90 \times 0.8^4$
 $= 36.864 \text{ cm}$

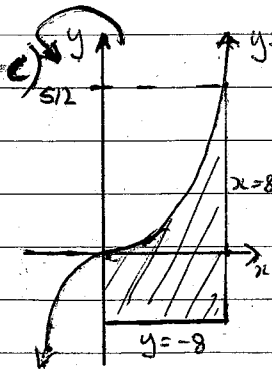
ii) $h = 120 + S_5$
 $= 120 + \frac{90(1-0.8^5)}{1-0.8}$
 $= 422.544 \text{ m}$

iii) $\text{Max } h = 120 + S_\infty$
 $= 120 + \frac{90}{1-0.8}$
 $= 570 \text{ cm}$
 $= 5.7 \text{ m}$

Question 6

a) $x^2 = 8(y-1) \quad a=2$
 i) $V(0,1)$
 ii) $S(0,3)$
 iii) $y=-1$

b) $(x-2)^2 + (y-3)^2 = (y+1)^2$
 $(x-2)^2 + y^2 - 6y + 9 = y^2 + 2y + 1$
 $(x-2)^2 = 8y - 8$

c)  $V = \pi r^2 h = \pi \int_0^{512} x^2 dy$
 $= \pi \times 8^2 \times 520 - \pi \int_0^{512} y^{\frac{2}{3}} dy$
 $= 33280\pi - \pi \left[\frac{3}{5} y^{\frac{5}{3}} \right]_0^{512}$
 $= 33280\pi - \frac{3\pi}{5} \left[512^{\frac{5}{3}} - 0^{\frac{5}{3}} \right]$
 $= 13619\frac{1}{5} \pi \text{ u}^3$
 $= 42786 \text{ u}^3$

Question 7

a) $(x+1)^2 + (y-2)^2 = (x-5)^2 + (y-4)^2$
 $x^2 + 2x + 1 + y^2 - 4y + 4 = x^2 - 10x + 25 + y^2 - 8y + 16$
 $12x + 4y = 36$

$3x + y = 9$

b) $x + y - 1 = 0$
 $y = 3 - x^2$
 $x + (3 - x^2) - 1 = 0$

$x^2 - x - 2 = 0$

$(x-2)(x+1) = 0 \quad \therefore x = 2, x = -1$
 $y = 3 - x^2$
 $y = 2, y = 2$

ii) $A = \int_{-1}^2 \left[(3 - x^2) - (1 - x) \right] dx$
 $= \int_{-1}^2 (-x^2 + x + 2) dx$
 $= \left[-\frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_{-1}^2$
 $= \left[\left(-\frac{8}{3} + \frac{4}{2} + 4 \right) - \left(\frac{1}{3} + \frac{1}{2} - 2 \right) \right]$
 $= \left(3\frac{1}{3} \right) - \left(-1\frac{1}{6} \right)$
 $= 4\frac{1}{2} \text{ u}^2$

Question 8

a) $P(x) = x^3 - 12x$

$P'(x) = 3x^2 - 12 \quad P'(x) > 0$

$3x^2 > 12$

$x^2 > 4$

$x < -2 \text{ or } x > 2$

b) $x + y = 24$

$P(x) = x^2 + (24 - x)^2$

$= 2x^2 - 48x + 576$

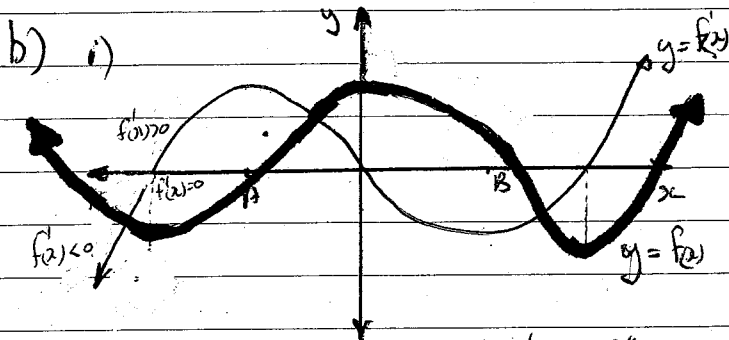
$P'(x) = 4x - 48 \quad P'(x) = 0, x = 12$

$P''(x) = 4 \quad \therefore \text{min at } x = 12, y = 12$

c) $I = \frac{1}{2} [\text{first} + \text{last} + 2 \times \text{others}]$
 $= \frac{0.5}{2} [1.3 + 0.6 + 2(1.8 + 2.1 + 1.2)]$
 $= 3.025$

Question 9

a) $(x-4)^2 + (y+3)^2 = 81$



ii) At A and B $y' = 0$, but $y' = f''(x)$
so $f''(x) = 0$ and y' changes direction
either side of A, B $\therefore$ concavity
changes on $y = f(x)$ at A and B.

$$\begin{aligned} \text{c) } A_1 &= 5000 \times 1.08^{20} \\ A_2 &= 5000 \times 1.08^{19} \\ &\vdots \\ A_n &= 5000 \times 1.08^n \\ S_n &= 5000 [1.08 + 1.08^2 + \dots + 1.08^{20}] \\ &= 5000 \left[\frac{1.08(1.08^{20} - 1)}{1.08 - 1} \right] \\ &= \underline{\underline{\$247114.61}} \end{aligned}$$

Question 10.

$$\begin{aligned} \text{a) } & (2^2 - 1) + (2^3 - 1) + (2^4 - 1) + (2^5 - 1) \\ &= 3 + 7 + 15 + 31 \\ &= \underline{\underline{56}} \end{aligned}$$

$$\begin{aligned} b) i) A_1 &= \$20\,000 - m \\ A_2 &= \$20\,000 - 2m \\ &\vdots \\ A_6 &= \$20\,000 - 6m \end{aligned}$$

$$\begin{aligned} \text{ii) } A_7 &= A_6 \times 1.015 - M \\ &= (\$20000 - 6m) \times 1.015 - m \end{aligned}$$

$$\begin{aligned} A_8 &= A_7 \times 1.015 - m \\ &= (\$20,000 - 6m) \times 1.015^2 - 1.015m - m \\ &= \underline{\underline{(20,000 - 6m) \times 1.015^2 - m(1 + 1.015)}} \end{aligned}$$

$$\text{iii) } A_{60} = (20000 - 6m) \times 1.015^{54} - m(1 + 1.015 + \dots + 1.015^{53})$$

$$\text{iv) } A_{60} = 0$$

$$\frac{20,000 \times 1.015^{54}}{c} = \frac{6 \times 1.015^{54} m}{(B \times m)} - \frac{m(1 + 1.015 + \dots + 1.015^{53})}{(m \times A)} = 0$$

$$\begin{aligned} \text{Now } 1 + 1.015 + 1.015^2 + \dots + 1.015^{53} &= S_n \text{ of GP.} \\ &= \frac{1(1.015^{54} - 1)}{1.015 - 1} \\ &= 82.295 \rightarrow A. (\text{memory}) \end{aligned}$$

$$6 \times 1.015^{54} = 13.41 \rightarrow B$$

$$20\,000 \times 1.015^{54} = 44\,688.55 \rightarrow C$$

So

$$C - B \times m - A \times m = 0$$

$$C = Bm + Am$$

$$= m(A+B)$$

$$m = \frac{c}{A+B}$$

467